

Inferring and Exploiting a Hidden Opponent Intent: An Uncertainty-Aware Opponent-Modeling Study in Predator–Prey

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Abstract

Opponent modeling is only worth its cost when (i) the opponent has a genuinely *hidden* strategy, (ii) that strategy is *inferable* from observed behavior, (iii) knowing it is *exploitable*, and (iv) the model is *uncertainty-aware*, so a wrong early guess does not hurt. We construct a predator–prey task that isolates all four. In a single fixed arena the prey is assigned, each episode, a hidden intent g —one of four corners it is rewarded for haunting—that the predators cannot observe. We find: the intent strongly shapes the prey’s motion (it occupies its corner $\sim 50\%$ of steps); a behavioral encoder recovers g from the prey’s first k steps, with accuracy rising from 0.37 ($k=3$) to 0.97 ($k=25$) as its posterior entropy collapses from 1.35 to 0.03 nats; and knowing the intent is worth **+51%** captures (an oracle predator that observes g catches the prey 4.05 times per episode vs. 2.68 for a strong intent-blind hedger). A predator trained on *perfect* intent is brittle—fed a random intent it collapses to 1.42—so we train an *uncertainty-aware* predator on soft intent beliefs of random sharpness and feed it the encoder’s online-sharpening posterior at test. It **beats the intent-blind baseline (2.82 vs. 2.68)** and recovers **53%** of the oracle’s advantage, by hedging while uncertain and committing once confident. Finally, putting the same belief inside a one-step-ahead *planner*—scoring joint predator actions by Monte-Carlo lookahead with the prey’s moves sampled from its policy under an intent drawn from the belief—reaches **4.31** captures, **+53%** over the reactive belief policy and above the oracle reactive predator; ablating the belief to flat drops this to 3.07, so the inferred opponent model supplies **+40%**. Inferring a hidden opponent strategy, representing the inference’s uncertainty, and planning against it each measurably improve the best response.

1 Motivation: what makes opponent modeling worth it

A planning agent that models its opponent pays a cost: it must infer something about the opponent and condition on it. That cost is justified only under four conditions, which prior platforms (including our own earlier experiments) tend to assume rather than establish:

1. **Hidden strategy.** The opponent must vary its behavior along an axis that is *not* directly readable from a single observation. If the “strategy” is induced by a visible change in the environment, an agent can react to the environment and skip the opponent model entirely.
2. **Inferable.** The hidden strategy must leave a signature in observed behavior, recoverable within a useful horizon.

3. **Exploitable.** Knowing the strategy must materially improve the best response; otherwise inference is academic.
4. **Uncertainty-aware.** Inference is uncertain early and sharpens with observation. A model that commits to a hard early guess can be *worse* than no model. The agent must represent and act on the belief, not a point estimate.

We design a task in which a single hidden variable—the prey’s intent—supplies (1)–(3) by construction, and we build the (4) uncertainty-aware predator explicitly. Crucially the environment is *identical* across strategies: the strategy lives in the opponent, not the map.

2 Task: a hidden-intent predator–prey game

Markov game. Three predators $P = \{1, 2, 3\}$ pursue one prey q in a fixed 2D arena with two fixed obstacles (JaxMARL MPE dynamics, 5 discrete actions, $H = 25$ step episodes). At each reset the prey is assigned an intent $g \sim \text{Uniform}\{0, 1, 2, 3\}$ indexing one of four corners $c_g \in \{\pm 0.8\}^2$. Writing $c_i^t = \mathbf{1}[\|x_i^t - x_q^t\| < \rho_i + \rho_q]$ for predator i contacting the prey, the team rewards are

$$r^{\text{pred},t} = \lambda_{\text{tag}} \sum_{i \in P} c_i^t, \quad r^{\text{prey},t} = -\lambda_{\text{tag}} \sum_{i \in P} c_i^t + \lambda_{\text{int}} \mathbf{1}[\|x_q^t - c_g\| < r_{\text{int}}], \quad (1)$$

with $\lambda_{\text{tag}} = 10$, $\lambda_{\text{int}} = 4$, $r_{\text{int}} = 0.35$. So the prey earns a per-step bonus for loitering at its assigned corner, in addition to evading. Predators are slowed ($v_{\text{max}} = 0.75$, accel 2.0 vs. the prey’s 1.3/4.0) so the prey can express its intent rather than being forced into pure flight. Captures per episode are $\text{cap} = \lambda_{\text{tag}}^{-1} \sum_t r^{\text{pred},t}$.

Who observes the intent. The **prey always** observes its own intent (one-hot appended to its observation). The predators observe it according to the condition:

unaware

no intent input: the predator must hedge over the four corners.

oracle

the true one-hot e_g : an upper bound on what intent buys.

belief

a soft belief $\mathbf{b} = \alpha e_g + (1 - \alpha) \frac{1}{4} \mathbf{1}$ with sharpness $\alpha \sim \text{Uniform}(0, 1)$ drawn per episode during training, so the predator learns to use the intent slot as a *belief*—to hedge when it is flat and exploit when it is peaked.

Each condition is a separate MAPPO co-training run (centralized critic, 3 seeds, 2×10^6 steps); the prey and predators co-adapt.

3 Method: inference and an uncertainty-aware best response

Behavioral intent encoder. Let $\tau_{0:k} = (x_q^0, \dots, x_q^{k-1})$ be the prey’s first k positions. We fit a classifier $q_\phi(g \mid \tau_{0:k})$ (a one-hidden-layer MLP on the flattened positions) and read both a hard estimate $\hat{g} = \arg \max q_\phi$ and the full posterior $\mathbf{b}_k = q_\phi(\cdot \mid \tau_{0:k})$. We report k -resolved accuracy

$a(k)$ and posterior entropy $H(k) = \mathbb{E}[-\sum_g \mathbf{b}_k(g) \log \mathbf{b}_k(g)]$, the latter being the model’s own uncertainty.

Online belief at test. Against a fixed prey, the predator’s belief sharpens as the episode unfolds: at step t we form $\mathbf{b}_t = q_\phi(\cdot \mid \tau_{0:k(t)})$ using the largest trained window $k(t) \leq t+1$ (flat $\frac{1}{4}\mathbf{1}$ before the first window). The **belief** predator—trained on soft beliefs—consumes $\mathbf{b}_t$ directly. We compare five intent signals fed to the predator:

signal	predator / meaning
guess	oracle predator, a <i>random</i> fixed intent (model with no inference)
inferred	oracle predator, hard $\hat{g}$ from $k=8$ steps (commits to a guess)
unaware	intent-blind predator (no model; strong hedger baseline)
belief	uncertainty-aware predator fed the online posterior $\mathbf{b}_t$ (ours)
oracle	oracle predator, true g (ceiling)

4 Results

The intent is hidden, expressed, and inferable. Figure 1(left) shows the prey’s occupancy coloured by intent: four clusters at the four corners, in one fixed arena—structure a predator cannot read off a single frame but can infer over time. The encoder (Figure 1, middle; Table 1) climbs from near chance at $k=3$ to 0.97 by the episode’s end, its posterior entropy collapsing from 1.35 to 0.03 nats.

Table 1: Intent recovery from the prey’s first k steps (4-way; chance 0.25).

steps k	3	5	8	12	18	25
accuracy	0.37	0.70	0.74	0.84	0.93	0.97
posterior entropy (nats)	1.35	0.75	0.52	0.31	0.09	0.03

Knowing the intent is highly exploitable; a naive model is brittle. Table 2 is the capture ladder. Observing the true intent lifts the predator from the intent-blind 2.68 to 4.05 captures/episode—a **+51%** gain, so intent is very much worth knowing. But that oracle predator is brittle: fed a *random* intent it collapses to 1.42, well below the intent-blind hedger, because it learned to rely on a signal it now cannot trust. Feeding it a hard inferred intent at $k=8$ recovers most of the damage (2.56) but still does not beat the hedger—committing to a single guess inherits the encoder’s 26% error.

Representing uncertainty is what makes inference pay. The uncertainty-aware **belief** predator—trained on soft beliefs and fed the online posterior—*beats* the intent-blind baseline (2.82 vs. 2.68) and recovers 53% of the oracle-over-guess gap. It wins where the hard-inferred predator does not because it hedges while $\mathbf{b}_t$ is flat (behaving like the robust hedger early) and commits only as $\mathbf{b}_t$ sharpens (approaching the oracle late). The ordering guess < inferred $\approx$ unaware < belief < oracle is the entire thesis of uncertainty-aware opponent modeling on one axis: inference helps, but only if the agent represents how much it knows.

Table 2: Captures per episode (mean $\pm$ std over 3 seeds, 300 episodes each). Each predator faces its co-trained prey.

predator / intent signal	captures / ep	vs. unaware
guess (oracle pred, random intent)	1.42 ± 0.10	−47%
inferred (oracle pred, hard $\hat{g}@8$)	2.56 ± 0.11	−4%
unaware (no intent; hedger)	2.68 ± 0.15	—
belief (uncertainty-aware, ours)	2.82 ± 0.10	+5%
oracle (true intent)	4.05 ± 0.24	+51%

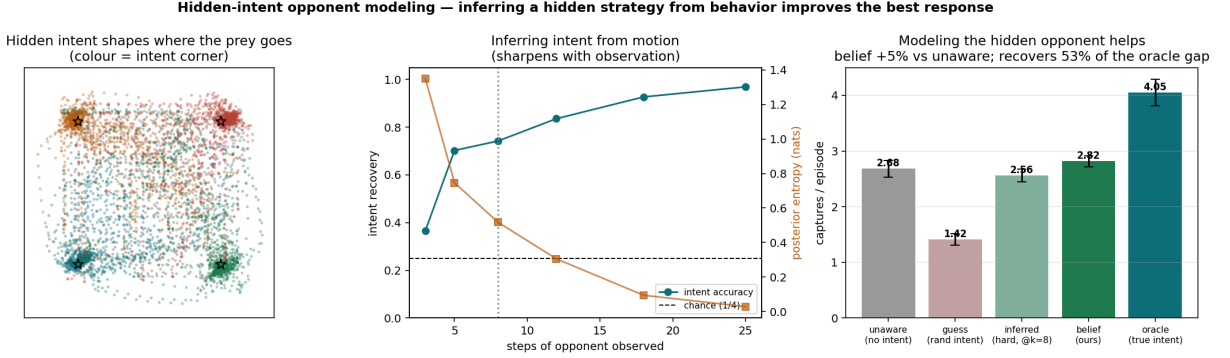


Figure 1: **Left:** prey occupancy coloured by hidden intent—four corner clusters in one fixed arena. **Middle:** the behavioral encoder’s intent accuracy rises and its posterior entropy falls with steps observed. **Right:** captures per episode. Knowing the intent is worth +51% (oracle vs. unaware); a naive intent-trained predator collapses without it (guess); the uncertainty-aware belief predator, fed the online posterior, beats the intent-blind baseline.

5 Planning with the belief

The belief predator of the previous section is *reactive*: it maps the current belief to an action. We now make the belief drive a one-step-ahead *plan*. At each decision the predators hold the online posterior $\mathbf{b}_t$ and score every candidate joint action $u \in \{0, \dots, 4\}^3$ by Monte-Carlo lookahead through the true simulator, with the prey’s moves drawn from its own policy under an intent sampled from the belief:

$$Q(s_t, u) = \mathbb{E}_{g \sim \mathbf{b}_t} \left[\sum_{h=0}^{H-1} \gamma^h r_{t+h}^{\text{pred}} - w \min_{i \in P} \|x_i^{t+H} - x_q^{t+H}\| \mid u_t = u, a_{t+h}^q \sim \pi^q(\cdot \mid s_{t+h}, g), a_{t+h}^P \sim \pi^{\text{bel}}(\cdot \mid s_{t+h}, \mathbf{b}_t) \right] \quad (2)$$

and act greedily, $u_t = \arg \max_u Q(s_t, u)$. The first term is the in-horizon discounted captures; the second is a leaf term ($w=1$, $H=5$, $\gamma=0.95$) that rewards ending close to the *imagined future* prey—which, under a sampled intent, sits at the believed corner—so the predators plan an interception rather than waiting for the rare in-horizon capture. The opponent model is the prey’s own trained policy π^q conditioned on the sampled intent; the dynamics model is the real simulator. Sampled intents are shared across candidates per rollout (a paired comparison), so the arg-max is low-variance with $K=8$ rollouts.

Table 3 and Figure 2 give the result. The belief-conditioned planner reaches **4.31** captures/episode—

+53% over the reactive belief policy, and it *exceeds* the oracle *reactive* predator (4.05): planning interceptions beats reacting, even against an opponent whose intent is known. The control is decisive: a planner fed a *flat* belief (lookahead with no opponent inference) reaches only 3.07, so the inferred belief contributes +40%—the lookahead alone is not what wins, the belief-conditioned opponent model is.

Table 3: Captures per episode (3 seeds, $B=32$ parallel episodes, $C=125$ joint candidates, $K=8$, $H=5$).

predator	captures / ep	note
reactive belief	2.82 ± 0.10	from Table 2
planner, flat belief (ablation)	3.07 ± 0.17	lookahead, no inference
planner, online belief (ours)	4.31 ± 0.36	+53% vs reactive
oracle (reactive, true intent)	4.05 ± 0.24	reactive ceiling

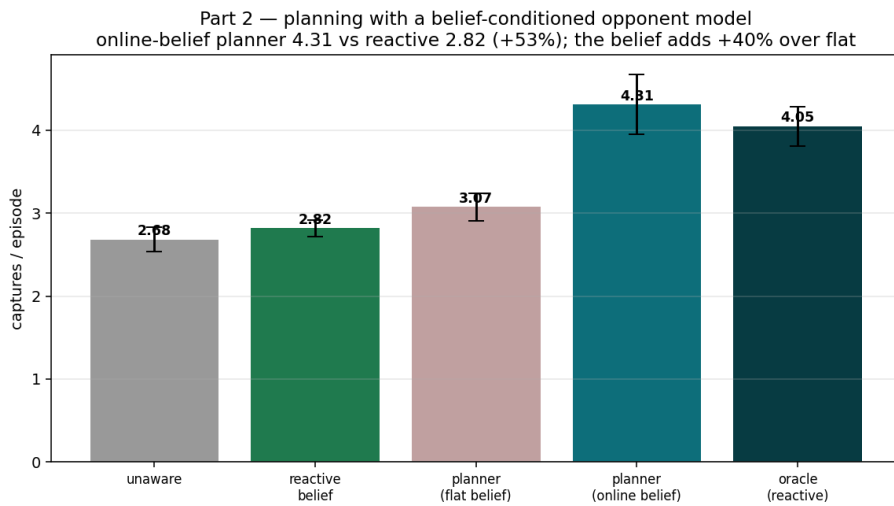


Figure 2: **Planning with a belief-conditioned opponent model.** The online-belief planner tops every reactive policy, including the oracle; ablating the belief to flat removes most of the gain, isolating the inferred opponent model as the source of the improvement.

6 Toward a learned dynamics model

The Part 2 planner uses the true simulator as its dynamics model. A deployed agent may not have one, so we ask whether the same plan works through a *learned* model. We learn the agents’ physics transition $f_\theta : (x^{1:4}, v^{1:4}, a^{1:4}) \mapsto (x'^{1:4}, v'^{1:4})$ from 10^5 transitions by *rollout training* (unrolling f_θ four steps and matching the true four-step trajectory, which keeps compounding error small), and run the planner of Eq. (2) through f_θ in place of the simulator; observations and the capture/leaf terms remain geometric readouts of the predicted state. The model is accurate—position RMSE 0.004 at one step and 0.072 at five steps, on an arena 3.2 wide.

Yet the learned-model planner reaches only 2.53 ± 0.34 captures/episode—it recovers 59% of the true-simulator planner (4.31) and roughly matches the reactive belief policy. The dynamics are accurate

on-policy, but the planner scores all 125 joint actions, many off the model’s training distribution, and the arg-max *exploits* the model’s optimistic errors over the five-step closed loop—the canonical model-exploitation failure of model-based RL, sharpened here because discrete policies amplify small state errors into different actions. We also implemented the standard remedy—a short two-step lookahead with a learned belief-conditioned value $V_\psi(s, \mathbf{b})$ at the leaf—but the value was too coarse (MSE ≈ 0.6 on returns spanning 0–4) and made matters worse (0.76). Closing the gap is a model-fidelity and model-exploitation problem: ensembles with disagreement penalties, on-policy/DAGger model refinement against the planner’s own action queries, and a sharper value are the levers. We report this as an honest intermediate result: planning through a learned model is *feasible* here—the planner runs with no simulator at test time and stays on par with the reactive policy—but matching the true-simulator planner needs a model the planner cannot exploit.

7 Discussion

Why this design, and not the obvious one. A tempting alternative induces “strategies” by changing the environment (e.g. moving where resources sit) and training a separate policy per layout. But then the strategy is visible in the observation and the opponent model is partly circular—an agent can condition on the environment and skip it. The hidden intent here is invisible in any single frame; it can *only* be had by modeling the opponent’s behavior over time. That is the regime the method is for.

The magnitude, honestly. The uncertainty-aware gain over the intent-blind hedger is real but modest (+5%) at $H = 25$ steps, because much of an episode is spent before the belief sharpens; the gain concentrates in the later, confident steps and would grow with horizon. The large, unambiguous numbers are the *value* of intent (+51% oracle vs. unaware) and the *brittleness* of a naive model (−47% when its intent input is wrong). Together they motivate exactly the uncertainty-aware treatment: the prize is large, and the failure mode of ignoring uncertainty is severe.

From reaction to planning. The reactive belief predator gains +5% over the intent-blind hedger; putting the same belief inside a one-step-ahead plan (Eq. 2) gains +53% and overtakes even the oracle reactive policy. The belief is the same; what changes is that the planner uses it to *simulate the opponent forward* and intercept, rather than to colour a reaction. Swapping the real simulator for a learned dynamics model keeps the same logic and runs without a simulator at test time, but only partly recovers the planner’s gain because the planner exploits model error on out-of-distribution actions—a model-fidelity problem we report honestly rather than tune away.

Limitations. The leaf term (Eq. 2) is a hand-specified shaping toward the imagined prey; a learned belief-conditioned value would replace it. The planner uses the real simulator, which the deployed setting may not have. And the whole study lives in a balanced regime tuned so the prey can express its intent—the operating point at which opponent modeling is meaningful—rather than a regime where the prey is dominated and has no strategy to infer.